MATH11219

LECTURE NOTES

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1 Differentiation

Differentiation (Derivatives) is just a mathematical expression of a functions rate-of-change. Think of the gradient of a function, area/estimation of a function, kinematics & optimisation of area or numbers.

1.1 Average Rate-of-Change (ARoC):

The ARoC of the function is just the estimation of a curve based on a linear lines gradient. I.E. the average gradient between two points.

$$\text{ARoC} = \frac{y_2 - y_1}{x_2 - x_1}$$

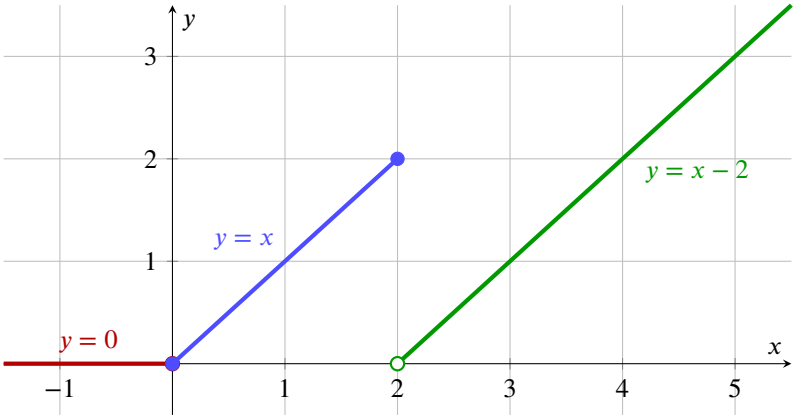
1.2 Instaneous Rate-of-Change (IRoC):

The ARoC of a function is good enough to understand the general direction in which a graph is continuing. Its accuracy can be improved by limiting the amount of change between x_1 & x_2. However a problem occurs when $\Delta x \to 0$. As cannot divide by 0. To find a functions IRoC, use the derivatives by first principles.

$$\begin{aligned}
f(x) &= 3x^2 + 2 \\
f'(x) &= \lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3(x + \delta x)^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3x^2 + 6x\delta x + 3\delta x^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{6x\delta x + 3\delta x^2}{\delta x} \\
&= \lim_{\delta x \to 0} 6x + 3\delta x \\
&= 6x
\end{aligned}$$

1.3 Limits in Functions:

Specifically in Piece-wise functions there can be an issue with derivatives and their interpretations. This is mostly around when the Left and Right Hand Limits not meeting on the same point, making the functions discontinuous.



As can be seem in the graph, at the point $x = 2$ there is not derivative as the function is discontinuous. However there moving in, up until that point a derivative is present. A similar story is true at the point $x = 0$. All other points along the x-axis have a derivative that exists.

1.4 Common Derivatives:

$y = f(x)$	$\frac{dy}{dx}$
c	0
x^n	$n \cdot x^{n-1}$
e^{kx}	$k \cdot e^{kx}$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{\ln(a) \cdot x}$
$\ln(x)$	$\frac{1}{x}$
$\sin(ax+b)$	$a\cos(ax+b)$
$\cos(ax+b)$	$-a\sin(ax+b)$
$\tan(ax+b)$	$a\sec^2(ax+b)$
$\csc(ax+b)$	$-a\csc(ax+b)\cot(ax+b)$
$\sec(ax+b)$	$a\sec(ax+b)\tan(ax+b)$
$\cot(ax+b)$	$-a\csc^2(ax+b)$
$\arcsin(ax+b)$	$\frac{a}{\sqrt{1-(ax+b)^2}}$
$\arccos(ax+b)$	$\frac{-a}{\sqrt{1-(ax+b)^2}}$
$\arctan(ax+b)$	$\frac{a}{1+(ax+b)^2}$
$\sinh(ax+b)$	$a\cosh(ax+b)$
$\cosh(ax+b)$	$a\sinh(ax+b)$
$\tanh(ax+b)$	$a\text{sech}^2(ax+b)$
$\text{csch}(ax+b)$	$-a\text{csch}(ax+b)\coth(ax+b)$
$\text{sech}(ax+b)$	$-a\text{sech}(ax+b)\tanh(ax+b)$
$\coth(ax+b)$	$-a\text{csch}^2(ax+b)$
$\text{arcsinh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2+1}}$
$\text{arccosh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2-1}}$
$\text{arctanh}(ax+b)$	$\frac{a}{1-(ax+b)^2}$

2 Techniques of Differentiation

2.1 Product Rule:

The product rule is used to differentiate two functions multiplied together.

$$\begin{aligned} y &= f(x)g(x) \\ \frac{dy}{dx} &= f'(x)g(x) + f(x)g'(x) \end{aligned}$$

$$\begin{aligned} y &= uv \\ \frac{dy}{dx} &= u'v + v'u \end{aligned}$$

2.2 Quotient Rule:

This rule is used when two functions divide each other. It is important to know that the order of functions matter with this rule. As this subtracts rather then addition.

$$\begin{aligned} y &= \frac{u}{v} \\ \frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \end{aligned}$$

2.3 Chain Rule:

By using differentiation by substitution (A.K.A) chain rule, function within other functions can be derived.

$$\begin{aligned} y &= f(g(x)) \\ \frac{dy}{dx} &= g'(x)f'(g(x)) \end{aligned}$$

2.4 Implicit Differentiation

This is used when the equation is not simply $y = f(x)$, e.g.:

$$x^2 - y^3 + \sin(x) - \cos(y) = 1$$

$$\begin{aligned}
\frac{d}{dx}(x^2) &= 2x \\
\frac{d}{dx}(-y^3) &= -3y^2\frac{dy}{dx} \\
\frac{d}{dx}(\sin(x)) &= \cos(x) \\
\frac{d}{dx}(-\cos(y)) &= \sin(y)\frac{dy}{dx} \\
\frac{d}{dx}(1) &= 0
\end{aligned}$$

$$\begin{aligned}
\frac{d}{dx}(x^2 - y^3 + \sin(x) - \cos(y)) &= \frac{d}{dx}1 \\
2x - 3y^2\frac{dy}{dx} + \cos(x) + \sin(y)\frac{dy}{dx} &= 0 \\
\frac{dy}{dx} &= \frac{-2x - \cos(x)}{\sin(y) - 3y^2} \\
&= \frac{2x + \cos(x)}{3y^2 - \sin(y)}
\end{aligned}$$

2.5 Parametric Differentiation

Basically just multiplying the derivatives together.

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

2.6 Logarithmic Differentiation

Just follow the basic steps and the derivative will be found. Mostly useful for a short version of the product rule.

1. Find the log of each side of the equation, & use log laws to expand the equation
2. Differentiate each side with respect too the independent variable
3. Rearrange and substitute
4. Simplify

For Example,

$$y = t^2(1+t)^3$$

$$\begin{aligned}\ln(y) &= \ln(t^2[1+t]^3) \\ &= \ln(t^2) + \ln([1+t]^3) \\ &= 2\ln(t) + 3\ln(1+t)\end{aligned}$$

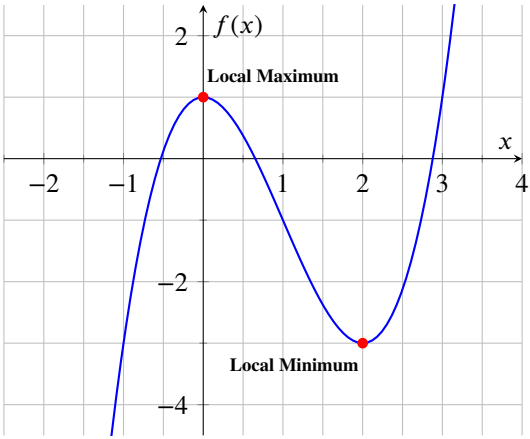
$$\begin{aligned}\frac{d}{dt}\ln(y) &= \frac{d}{dt}[2\ln(t) + 3\ln(1+t)] \\ \frac{1}{y}\frac{dy}{dt} &= \frac{2}{t} + \frac{3}{1+t} \\ \frac{dy}{dt} &= y(\frac{2}{t} + \frac{3}{1+t})\end{aligned}$$

2.7 Higher Derivatives

Found by differentiating the function multiple times.
These are useful as they can describe the nature of a function, such as
If $\frac{dy}{dx} > 0$, the gradient can be said to be increasing, or $\frac{dy}{dx} < 0$, the gradient is decreasing and $\frac{dy}{dx} = 0$ the gradient is stable.
Or $\frac{d^2y}{dx^2} > 0$, concavity is up and the point (x,y) is described as a minimum. $\frac{d^2y}{dx^2} = 0$, described as a Point of Inflection. Basically just the point where the concavity of a function changes. $\frac{d^2y}{dx^2} < 0$, concave down and point (x,y) is a maximum.

3 Applications of Derivatives

3.1 Maximum & Minimum Points



These points are found by substituting the x-values found when $\frac{dy}{dx} = 0$ into $\frac{d^2y}{dx^2}$. In addition POIs can be found when $\frac{d^2y}{dx^2} = 0$. A simple way for remembering which point is what is understanding how the gradient is achieved around the point.

3.2 Points of Inflextion (POI)

This is the point in which the graph changes in concavity.
Maximum - Concave Down
Minimum - Concave Up

3.3 Newton-Raphson Method

This is an estimation method used to find roots within an equation. The reason for this is for polynomials not able to be found by hand. For the method to work it assumes point x_2 is closer too the root value than x_1. However the method can come off course as it is entirely dependent upon the intial value chosen to represent point x_1. As this can cause the approximation to diverge (rather than converge) upon a particuar point.

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$

4 Sequences & Series:

4.1 Sequences

A sequence is a set of numbers in a particular order with most cases following a pattern.
E.g. $x[k] = (-1)^{k+1} \cdot \frac{1}{k},\ \forall k \geq 1$ becomes,
$1, \frac{-1}{2}, \frac{1}{3}, \ldots$

4.2 Series

Series are the sum of multiple sequences up until a certain point, eg.

$$\sum_{n=1}^{5}(2n-1) = 1+3+5+7+9$$

4.3 Convergence

If an infinite series is said to converge upon a value if and only if it tends towards a singular value. For example $\lim_{k\to\infty}\ \frac{1}{k}$ converges towards 0. A sequence can converge on a non-zero quantity. For a point that might be considered to converge on ∞ is instead called to diverge. E.g. $\lim_{k\to\infty} \sqrt{k} = \infty$.

$$x[k] = \frac{3k^2+2k-7}{9k^2-7k},\ k \to \infty$$

$$\begin{aligned}
&\lim_{k\to\infty} \frac{3k^2+2k-7}{9k^2-7k} \\
=&\lim_{k\to\infty} \frac{(3k^2+2k-7)\times\frac{1}{k^2}}{(9k^2-7k)\times\frac{1}{k^2}} \\
=&\lim_{k\to\infty} \frac{3+\frac{2}{k}-\frac{7}{k^2}}{9-\frac{7}{k}} \\
=&\frac{3}{9}\text{ , because } \lim_{k\to\infty}\frac{a}{k} = 0 \\
=&\frac{1}{3}
\end{aligned}$$

4.4 Arithmetic Sequences

This is a type of sequence in which each values difference is the same amount.

$$a_n = a_1 + (n-1)d$$

These vaues can also be summed to a total series. The proof of this is beautiful thing to see, look it up online.

$$S_n = \frac{n}{2}(2a_1 + [n-1]d)$$

4.5 Geometric Sequences

This a sequence of numbers related to a common ratio of r.

$$a_n = a_1 \cdot r^{n-1}$$

$$S_n = \frac{a_1(1-r)}{1-r},\ r \neq 1$$

If $-1 \leq r \leq 1$,

$$S_n = \frac{a_1}{1-r}$$

This is also the only one that can find S_∞. Outside of $-1 \leq r \leq 1$ r^n fails to converge on a finite limit as $n \to \infty$, thus the series and sequence diverges.

4.6 Power Series

$$\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\
\sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \approx x \\
\cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \approx 1 - \frac{x^2}{2!} \\
\log(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots,\ -1 \leq x \leq 1
\end{aligned}$$

4.7 Iterative Solutions to Non-linear Equations

$$\begin{aligned}
f(x) &= e^{-x} - x = 0 \\
\therefore x &= e^{-x} \\
x_{n+1} &= e^{x_n}
\end{aligned}$$

5 Taylor & Maclaurin Series

5.1 Approximating Points on a Function

Think amount the point (Q) on a function. When an approximation of the function is close to Q it is considered to be fairly accurate. However as the distance between the approximation and Q grows.

$$f(x) = f(a) + f'(a)(x - a)$$

5.2 Taylor Series

Taylor Series becomes more accurate the higher order (highest power in the set) is.
Zero: $p_0(x) = f(a)$
First: $p_1(x) = f(a) + f'(a)(x - a)$
Second: $p_2(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2$
Continues with higher orders

5.3 Remainder Term

This is the error in the nth term of a Taylor Polynomial.

$$f(x) = p_n(x) + R_n(x)$$

$$R_n(x) = \frac{y^{n+1}(c)(x - a)}{(n + 1)!}, \text{ where } a < c < x$$

5.4 Maclaurin Series

Think of this as a Taylor polynomial, except that it rotates around the point $x = 0$.

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!} + \frac{f^{(3)}(0)}{3!}x^3 + \ldots$$